Exercises

1. In 1D, Newton's method becomes unstable when the second derivative f''(x) is close to 0 because the update step requires dividing by this value. In multidimensional optimization, the 2nd derivative is replaced by the Hessian matrix. The analog to f''(x)≈0 is when the Hessian matrix becomes singular. This occurs when the determinant of the Hessian is approaching 0. In this state, the curvature of the function is flat in at least 1 direction. This makes it impossible to calculate the Newton step reliably.

Computerlab

1) Part a,b,c:

```
--- Part (a): Golden Section Search ---
Maxima found: 1.18715127
Iterations: 31
Error: 1.27e-06

--- Part (b): Newton's Method ---
Maxima found: 1.18715144
Iterations: 4
Error: 1.44e-06

--- Part (c): Quadratic Interpolation ---
Maxima found: 1.18715140
Iterations: 8
Error: 1.40e-06
```

Part D: Golden Search required the most iterations.

Newton's Method is extremely fast, provided the initial guess is decent and derivative is derived correctly.

Quadratic Interpolation falls in between with being faster than golden but slower than Newton’s.

2) A:

Comp lab 2

9 $f(x_1, x_2) = (x_1 + 2x_2 - 7)^2 + (2x_1 + x_2 - 5)^2$

Gradiant

$u = x_1 + 2x_2 - 7$

$v = 2x_1 + x_2 - 5$

$f = u^2 + v^2$

$$\frac{\partial f}{\partial x_1} = 2u \frac{\partial u}{\partial x_1} + 2v \frac{\partial v}{\partial x_1} = 2(2x_1 + x_2 - 5)(2) + 2(x_1 + 2x_2 \ldots$$

$= 10x_1 + 8x_2 - 34$

$$\frac{\partial f}{\partial x_2} = 2u \frac{\partial u}{\partial x_2} + 2v \frac{\partial v}{\partial x_2} = 2(x_1 + 2x_2 - 7)(2) + 2(2x_1 + x_2 - 5)(\ldots$$

$= 8x_1 + 10x_2 - 38$

$$\nabla f = \begin{bmatrix} 10x_1 + 8x_2 - 34 \\ 8x_1 + 10x_2 - 38 \end{bmatrix}$$

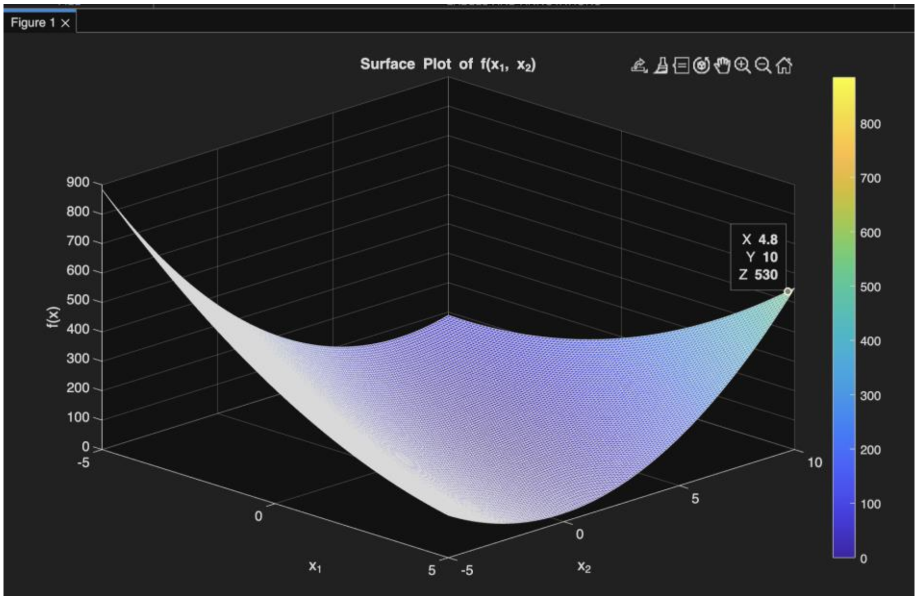
Hessian

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x_1 + \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 10 & 8 \\ 8 & 10 \end{bmatrix}$$

crit point

$5x_1 + 4x_2 = 17$ $\quad x_1 = 1$

$4x_1 + 5x_2 = 19$ $\quad x_2 = 3$ $\quad$ crit point $= (1, 3)$

B: 

C and D:

```
--- Part (c): Newton's Method ---
Start Guess: [0.0, 0.0]
  Minima: [1.000000, 3.000000]
  Iterations: 2
Start Guess: [5.0, 5.0]
  Minima: [1.000000, 3.000000]
  Iterations: 2

--- Part (d): BFGS Method ---
Start Guess: [0.0, 0.0]
  Minima: [1.000000, 3.000000]
  Iterations: 5
Start Guess: [5.0, 5.0]
  Minima: [1.000000, 3.000000]
  Iterations: 5
```

E: Newton's method converges in 1 iteration (excluding the check step). This is because the objective function is purely quadratic, so the Quadratic approximation used by Newton's method is exact. BFGS takes more iterations because it iteratively builds the Hessian approximation. However, BFGS eventually converges to the same result without calculating exact second derivatives.

AI Prompts used

Computer Lab 1:

Write a MATLAB script to solve the following 1D optimization problem.

Objective:

Maximize the function f(x)=1−(x−1)2+sin(x) with a tolerance of ϵ=10−6. The function is unimodal in the interval [0,3].

Requirements:

Golden Search Method: Implement this method using the bracket [0,3]. Output the maximum found and the number of iterations required.

Newton's Method: Implement this method starting with x0 =0. You will need to analytically derive the first and second derivatives of f(x). Output the maximum found and the iteration count.

Quadratic Interpolation: Implement this method starting with points x0 =0, x1 =1.5, and x2 =3. Output the maximum found and the iteration count.

Computer Lab 2

Write a MATLAB script to solve the following multidimensional optimization problem.

Objective:

Minimize the function f(x1 ,x2)=(x1 +2x2 −7)2+(2x1 +x2 −5)2.

Requirements:

Define Derivatives: Define the function, its Gradient, and its Hessian matrix in the code (derive these analytically first).

Plotting: Use surf to plot the function. Center the domain around the critical point (which you should calculate or approximate as roughly (1,3)).

Newton's Method: Implement Newton's method for multidimensional optimization to find the minimum. Run it twice using two different initial guesses. Use a tolerance of 10−6.

BFGS Method: Implement the BFGS quasi-Newton method to find the minimum. Run it twice using the same two initial guesses as above. Use a tolerance of ϵ=10−6 and a maximum iteration limit of 50.

Output: For each method and guess, print the calculated minimum location and the number of iterations required.